International Journal of Fatigue 188 (2024) 108535



Contents lists available at ScienceDirect

International Journal of Fatigue

journal homepage: www.elsevier.com/locate/ijfatigue





Neural network integrated with symbolic regression for multiaxial fatigue life prediction

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ARTICLE INFO

Keywords:
Multiaxial fatigue
Life prediction
Symbolic regression
Neural network
SR-NN

ABSTRACT

In the realm of structural engineering, accurately predicting multiaxial fatigue life presents a formidable challenge, stemming from the complex interplay of stress and strain across multiple directions. This research introduces the Symbolic Regression-Neural Network (SR-NN) framework, a novel integration of symbolic regression-derived expressions with neural networks aimed at enhancing predictive accuracy in this field. Initiated by selecting salient features through SHAP-informed Recursive Feature Elimination, our approach effectively minimized dimensionality while pinpointing key feature groups. Symbolic regression, newly applied to guide neural network training in the prediction of alloy material fatigue life, facilitated the generation of meaningful expressions that describe the fatigue processes. These expressions were incorporated into the network's loss function alongside MSE loss, refining the model's learning dynamics. Our proposed SR-NN framework demonstrated substantial improvements over traditional models in rigorous testing, notably reducing MSE by 55.4% compared to RF and 45% against SVM models, underscoring the potential of merging symbolic regression with machine learning to tackle the inherent variability and complexity of fatigue behavior across diverse materials and loading conditions. This study highlights the transformative impact of symbolic regression in improving the interpretability and accuracy of predictive models within structural engineering.

1. Introduction

Multiaxial fatigue frequently stands as a principal concern in structural engineering[1]. Understanding and predicting the fatigue life of materials under complex stress states is a cornerstone challenge in the fields of materials science and engineering[2]. Multiaxial fatigue, characterized by the presence of stress components in multiple directions, complicates the prediction of fatigue life due to the complex interactions of stress and strain within the material. The nature of real-world applications often subjects materials to multiaxial loading conditions, making it imperative to develop accurate predictive models that can account for these complexities.

Indeed, empirical formulas derived from rigorous laboratory testing have long been the standard for fatigue life estimation[3,4]. Models based on the critical plane approach[5] or energy-based[6] frameworks stand as exemplars. According to these theories, particular forms of stress[7], strain[8], or energy[9] are elected as the primary factors driving fatigue. Such variables are thus prominently featured in the resultant mathematical expressions. However, it is worth noting that the application of these models is typically restricted to certain material types, loading scenarios, and stress levels, underscoring the necessity for versatile methodologies that can encompass a broader spectrum of real-world conditions.

Over the past few years, machine learning has been gaining ground in the field of fatigue analysis[10–17]. These approaches differ from traditional models, which often focus on broad phenomena and accumulated empirical findings. Instead, machine learning concentrates on understanding the specific connections between input data and their outcomes. This method allows for more nuanced interpretations of how different factors contribute to fatigue, providing a more direct route from data to predictions. Recent studies have explored various machine learning techniques in multiaxial fatigue life prediction, including artificial neural networks[18], branched neural network structures[19], Bayesian neural networks[20], hybrid models combining convolutional neural networks and long short-term memory (LSTM)[21], and deep learning models with self-attention[22]. Comparative studies have shown that these machine learning methods often outperform

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https://doi.org/10.1016/j.ijfatigue.2024.108535
Received 19 April 2024; Received in revised form 8 July 2024; Accepted 28 July 2024
Available online 29 July 2024
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Nomenclature

E	Tensile Modulus of Elasticity
K'	Fatigue Strength Coefficient
n'	Fatigue Strength Exponent
σ'_f	Cyclic Hardening Coefficient
ε'_f	Fatigue Ductility Coefficient
b	Fatigue Ductility Exponent
c	Cyclic Hardening Exponent
G	Shear Modulus of Elasticity
K'_0	Shear Fatigue Strength Coefficient
n'_0	Shear Fatigue Strength Exponent
τ'_f	Shear Fatigue Ductility Coefficient
γ'_f	Shear Fatigue Ductility Exponent
b_0	Shear Cyclic Hardening Coefficient
c_0	Shear Cyclic Hardening Exponent
P_1	Path-related Parameter
P_2	Path-related Parameter
layer size	neurons per layer
alpha	regularization strength
learning rate	optimization step size
activate function	neuron activation mechanism
c	penalty parameter
kernel	data transformation function
degree	polynomial kernel degree
gamma	kernel coefficient
n_estimators	number of trees
max depth	tree depth limit
min samples split	minimum samples for node split
min samples leaf	minimum samples at leaf node
subsample	training instance fraction
colsample bytree	feature sampling ratio

traditional empirical models in terms of prediction accuracy[23]. This superior performance can be attributed to their ability to capture complex, non-linear relationships and interactions among multiple variables simultaneously, without being constrained by predefined functional forms or simplifying assumptions inherent in many traditional models.

Despite these advancements, the application of machine learning in fatigue analysis faces a significant challenge: the scarcity of comprehensive datasets. Due to the high cost and time-consuming nature of fatigue testing, particularly for multiaxial tests, the availability of data points is often limited. This scarcity poses difficulties in training robust ML models, as those trained on insufficient datasets often demonstrate poor performance when applied beyond the scope of the training data [24]. To address this issue and enhance the generalizability of ML models, researchers have begun to incorporate physical laws into machine learning, giving rise to Physics Informed Neural Networks (PINN) in fatigue life prediction[25]. This approach aims to leverage the strengths of both data-driven methods and physical understanding.

Various strategies have been employed to integrate physical knowledge into machine learning models for fatigue life prediction. One methodology directly incorporates physical models into neural networks [26,27]. For instance, He et al. [28] integrated multiaxial fatigue critical plane analysis into their physics-informed neural network model, and subsequently[29] employed physics-informed GANs to generate synthetic data, enhancing the model's predictive capabilities under varied loading conditions. The alternative methodology utilizes physical principles to constrain the neural network architecture[30]. Zhou et al. [31] exemplified this approach by imposing constraints based on the Basquin-Coffin-Manson equation in their ANN model, effectively embedding fundamental fatigue principles into the learning process. These physics-informed approaches have demonstrated significant potential in enhancing model accuracy and robustness, particularly in scenarios involving limited datasets or extrapolation to novel conditions.

The integration of PINN with established physical laws represents a leap forward in fatigue life prediction. However, the reliance on empirical formulas, stemming from material-specific empirical data and stringent requirements on material attributes. Empirical formulas are not selected by identification of the physical damage mechanism, but rather the best performing model is determined based on the fatigue life prediction accuracy[23]. While they can guide machine learning methods to higher precision, their application is often fraught with challenges due to their tailored nature, potentially limiting generalizability and hindering model accuracy when applied inappropriately. Given the variability of fatigue behavior with respect to material, shape, and conditions, compounded by the emergence of novel materials in recent years, it becomes particularly challenging to develop a standardized model for predicting fatigue life[32]. This has prompted a pivot towards mining the inherent features of datasets to aid predictions [28].

Despite advancements in physics-constrained neural networks, the complexity of fatigue behavior suggests potential for undiscovered patterns beyond current physical models. Exploring complementary methodologies that leverage intrinsic dataset information could yield additional insights into fatigue mechanisms. In this context, symbolic regression stands out as a technique that leverages the inherent patterns in data, circumventing the constraints of traditional formulas and offering a path to more universal predictive modeling[33–37]. This method aligns with the goal of deriving insights directly from the data, potentially revealing relationships that might be obscured by preconceived physical models. Among the relatively rare works, Gan et al. [38] employed symbolic regression to propose an interpretable nonlinear damage accumulation model for predicting fatigue under multistep loading. The symbolic regression can also generate expressions as additional nonlinear features for the training of machine learning models[39,40]. This introduces more nonlinear information into the machine learning model training process, ultimately leading to improved model performance. In spite of the work mentioned above, it is noted that in the domain of life prediction, applications of symbolic regression are predominantly confined to utilizing the derived expressions either directly for prediction purposes or merely incorporating the results of symbolic regression as new input features. Although symbolic regression can generate human-readable expressions to enhance understanding of machine learning models, to the best of the authors' knowledge, there has been no research combining symbolic regression with machine learning to guide the training of machine learning algorithms for improving fatigue life prediction performance.

To address these concerns, this study proposed the Symbolic Regression-Neural Network (SR-NN) framework, an innovative approach that significantly improves the predictive accuracy of machine learning models in the context of multiaxial fatigue life analysis. At the outset, a rigorous feature analysis is conducted to identify key feature groups. Through symbolic regression, insightful expressions that reflect the physical dynamics between key features and multiaxial fatigue life are extracted. These foundational expressions are seamlessly integrated into the architecture of artificial neural networks, significantly boosting their predictive precision.

Section 2 introduces the database utilized and outlines our newly proposed SR-NN method. Section 3 presents the results of our predictions, showcasing the efficacy of the proposed method in accurately predicting multiaxial fatigue life. Finally, the main findings and

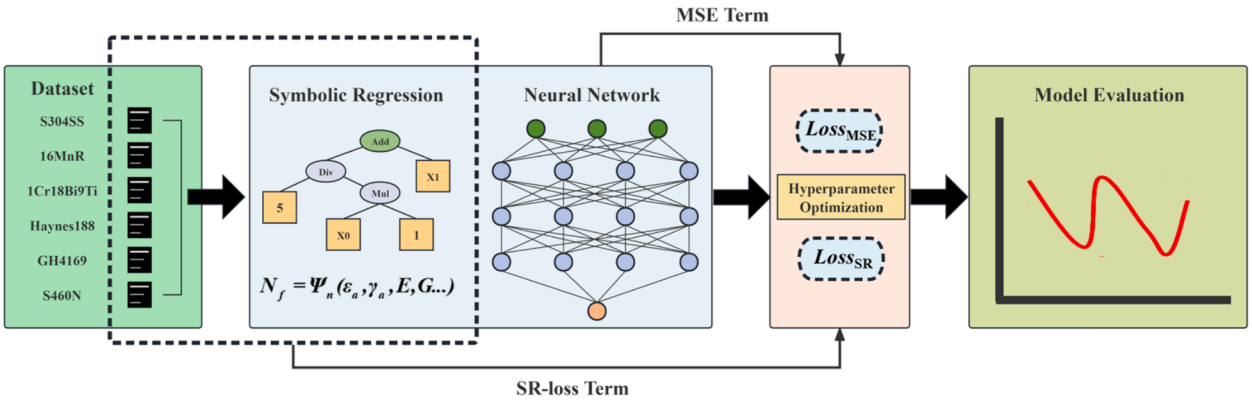


Fig. 1. Schematic of the Prediction Framework Incorporating Symbolic Regression and Neural Networks. The process begins with data input, followed by symbolic regression to derive expressions that inform the neural network, which is then refined using MSE and symbolic regression loss terms leading up to the model testing phase.

Table 1
Fatigue properties of the materials used in this paper.

Materials	304SS	16MnR	1Cr18Bi9Ti	Haynes188	GH4169	S460N
E(GPa)	185	212.5	193	170.2	182	208.5
K'(MPa)	1660	1106	1115	891	1950	1115
n'	0.287	0.186	0.1304	0.113	0.15	0.161
σ_f'(MPa)	1000	966.4	1124	823	1565.2	834
ε_f'	0.171	0.842	0.8072	0.489	0.162	0.1572
b	−0.114	−0.101	−0.0905	−0.082	−0.086	−0.079
c	−0.402	−0.618	−0.6652	−0.73	−0.58	−0.493
G(GPa)	82.8	81.1	74.3	64.4	70	80.2
K_0'(MPa)	785	535.5	1053	589	1037	753.8
n_0'	0.296	0.15	0.13	0.142	0.153	0.23
τ_f'(MPa)	709	886	644	635	904	529
γ_f'	0.413	1.562	0.812	1.78	0.281	0.213
b_0	−0.121	−0.105	−0.0876	−0.1	−0.086	−0.096
c_0	−0.353	−0.648	−0.5334	−0.715	−0.58	−0.418

contributions are summarized in section 4.

2. Methodology

This section outlines the proposed methodology, starting from dataset preparation, progressing through feature analysis, and culminating in the implementation of a novel prediction framework. Fig. 1 provides a panoramic view of this framework, which integrates inputs from datasets into a combined approach of symbolic regression and neural network architecture, leveraging both traditional and symbolic regression-derived loss functions for thorough model testing.

2.1. Database establishment

2.1.1. Experimental data acquisition

The materials used in this work have been obtained from a number of sources[41–50]. Multiaxial experimental data of six materials were generated in strain-controlled mode. The dataset contains features organized under two principal categories: the fatigue loading condition, and aspects related to the mechanical property, as detailed in Table 1. These features include mechanical properties, fatigue characteristics, and material constants that describe the stress–strain behavior under cyclic loading. This study primarily focuses on the influence of loading conditions and material-related parameters on fatigue life prediction, considering various strain states and cyclic material properties.

Due to the varying load paths collected in the dataset, to simplify the input features for the neural network, this study employs a parameter-based description of the load paths, specifically using the rotational inertia method to introduce two path-related parameters, P_1 and P_2

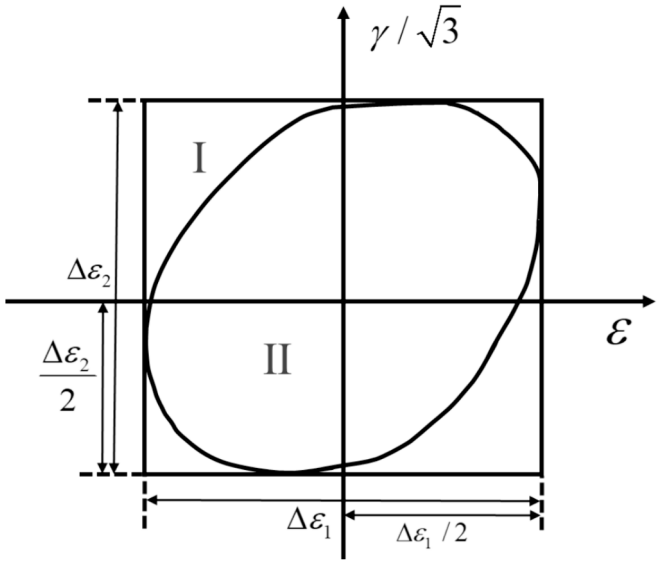


Fig. 2. Schematic of Load Path and Strain Range. The rectangular dimensions $\Delta\varepsilon_1$ and $\Delta\varepsilon_2$, represent the ranges of strain and shear strain, with the irregular curve denoting the actual load path.

[9,17]. As shown in Fig. 2, the load path and the strain range axial strain $\Delta\varepsilon_1 = \varepsilon_{max} - \varepsilon_{min}$ and torsional strain $\Delta\varepsilon_2 = (\gamma_{max} - \gamma_{min})/\sqrt{3}$ form an irregular load path curve. The rectangular part formed by $\Delta\varepsilon_1$ and $\Delta\varepsilon_2$ is

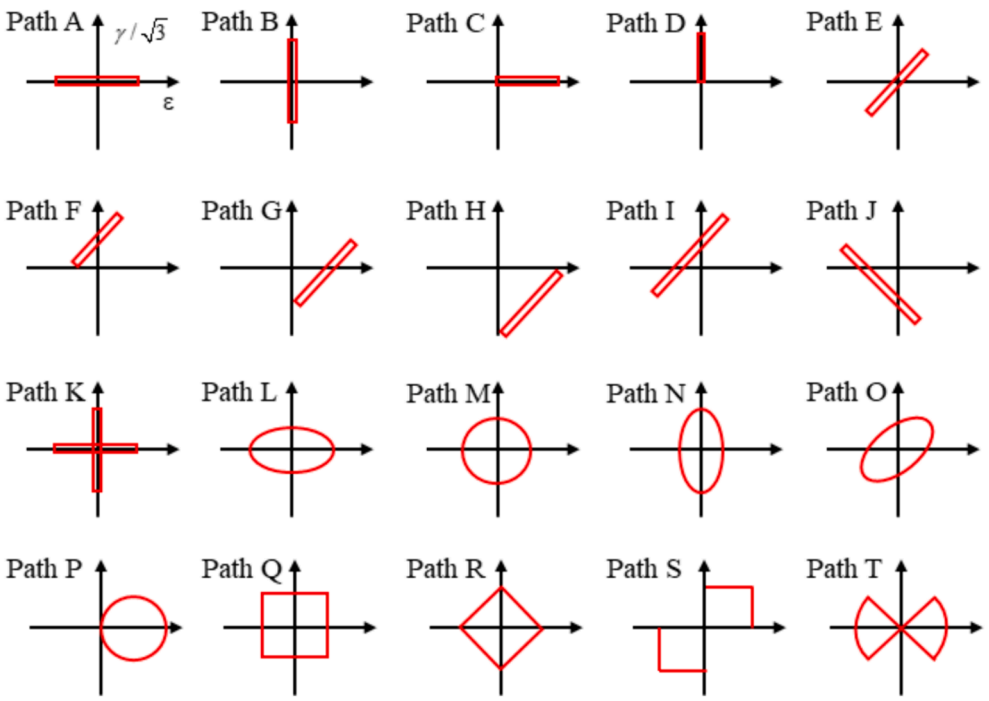


Fig. 3. Schematic Representation of Stress-Strain Loading Paths. Paths A to T depict different modes of loading, including unidirectional and complex multiaxial sequences. In this context, uniaxial loading is considered a specific case of multiaxial loading conditions.

Table 2
Correspondence of Materials to Their Respective Stress-Strain Loading Paths.

Materials	Loading Path
S304SS	A, B, E, M, P, R
16MnR	A, B, E, I, J, M
1Cr18Bi9Ti	A, B, M
Haynes188	E, M, N
GH4169	E, L
S460N	A, B, E, F~H, K, M, N~Q, S, T

denoted as I_1, while the irregular curve path area is denoted as I_2. The area moment of inertia for I_1 with respect to the axial strain ε axis is denoted as I_{11}, and with respect to the torsional strain $\gamma/\sqrt{3}$ axis is denoted as I_{12}. Similarly, the area moment of inertia for I_2 with respect to the ε axis is denoted as I_{21}, and with respect to the $\gamma/\sqrt{3}$ axis is denoted as I_{22}. Therefore, the introduced path influence parameters P_1 and P_2 are defined as:

$$P_1 = \frac{I_{11}}{I_{12}} \tag{1}$$

$$P_2 = \frac{I_{22}}{I_{12}} \tag{2}$$

P_1 reflects the influence of the change in γ on the fatigue life during the loading process, while P_2 reflects the influence of the change in ε on the fatigue life during the loading process. Here, the load path P can be transformed into the form of path M through translation, and the corresponding P_1 and P_2 can be obtained. It is worth noting that for uniaxial load paths and proportional load paths, P_1=0 and P_2=0, for path Q, P_1=1 and P_2=1, for path Q, P_1=0.5 and P_2=0.5. Fig. 3 shows a graphical representation of the load paths involved. The loading paths associated with each material are delineated in Table 2. It is observable that S460N encompasses the widest range of loading paths, whereas GH4169 is associated with the fewest. Fig. 4(a) presents the proportion of various materials within the database, while Fig. 4(b) displays the distribution of

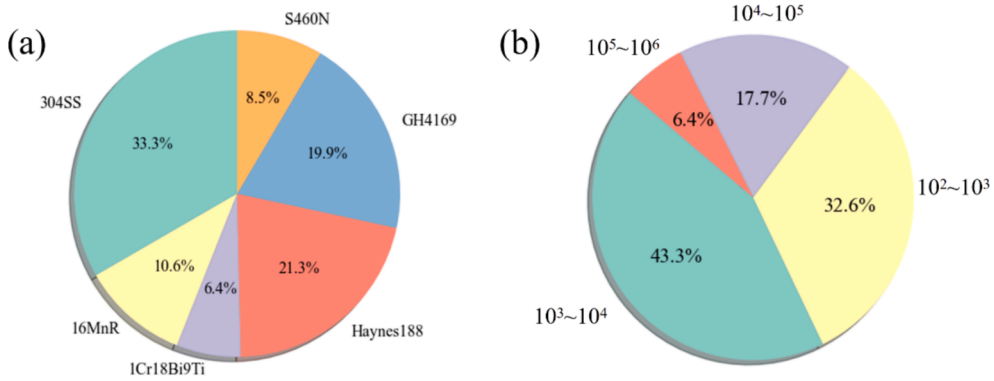


Fig. 4. Comparative Analysis of Material Composition and Lifespan Distribution. (a) illustrates the percentage share of each material type in the database, highlighting the prevalence of 304SS as the most significant. (b) shows the lifespan categories of the components, with a notable majority falling within the 10^3 to 10^4 range.

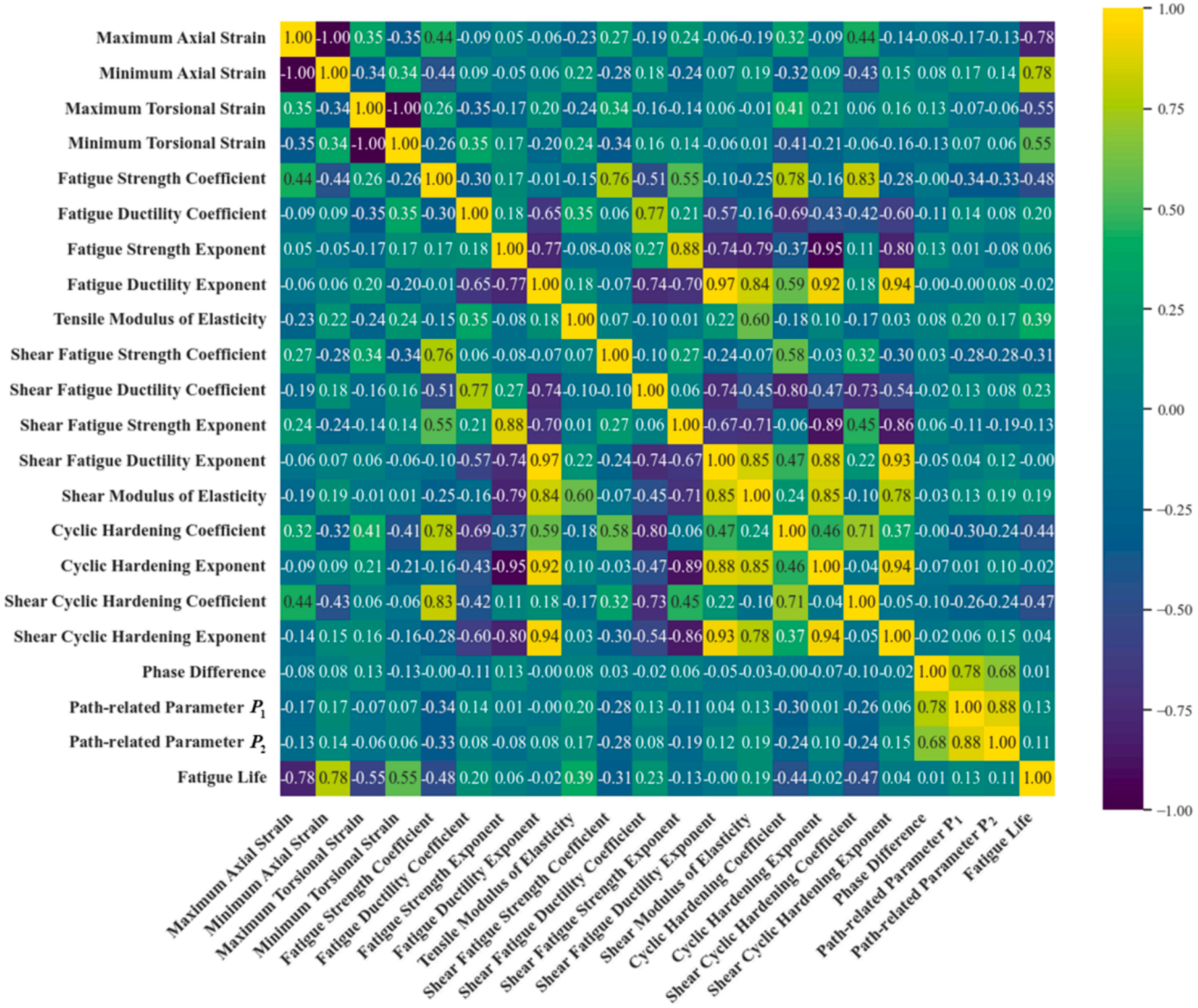


Fig. 5. Pearson Correlation Coefficient Heatmap Depicting Linear Associations between Material Properties. The color scale from purple to yellow represents correlation values ranging from −1 (perfect negative correlation) to 1 (perfect positive correlation), facilitating the visualization of significant correlations amongst the dataset features.

different lifespans.

Overall, the dataset is broadly characterized, containing 133 different data sets covering a wide range of mechanical properties and stress–strain scenarios. The richness and diversity of the data makes it particularly valuable for machine learning algorithms that can focus on accurately predicting fatigue life under different operating conditions.

2.1.2. Data preprocessing

The standardization of datasets is an essential step prior to the model training phase, serving to nullify the disparity in the scale of attributes and to expedite the convergence of iterative algorithms. This process is implemented through a range transformation method, which is delineated as follows:

$$X_N = \frac{X_i - X_{\min}}{X_{\max} - X_{\min}} \tag{3}$$

where X_i denotes the original value of a given attribute, $X_{\min}$ and $X_{\max}$ signify the minimum and maximum values observed for that attribute across the dataset, respectively. The variable X_N is the standardized result, representing the attribute value adjusted to a uniform scale. The process of normalization aims to eliminate dimensional discrepancies among features, thereby speed up convergence speed while enhancing the stability and performance of predictions. The output, which is the predicted fatigue life N_f, undergoes logarithmic transformation for scaling purposes, effectively reducing the span of the data.

2.2. Feature analysis

To prepare for symbolic regression by analyzing a multitude of features and selecting key feature groups, we have employed Pearson correlation coefficient (PCC) analysis, Recursive Feature Elimination (RFE) based on the Random Forest (RF), and the SHapley Additive exPlanations (SHAP) method.

2.2.1. Pearson correlation coefficient analysis

The PCC[51] is employed as a metric to gauge the degree of association between features. A higher PCC value indicates an increased level of correlation between the variables in question. The PCC is calculated as the ratio of the covariance of two variables to the product of their individual standard deviations, as delineated in Equation:

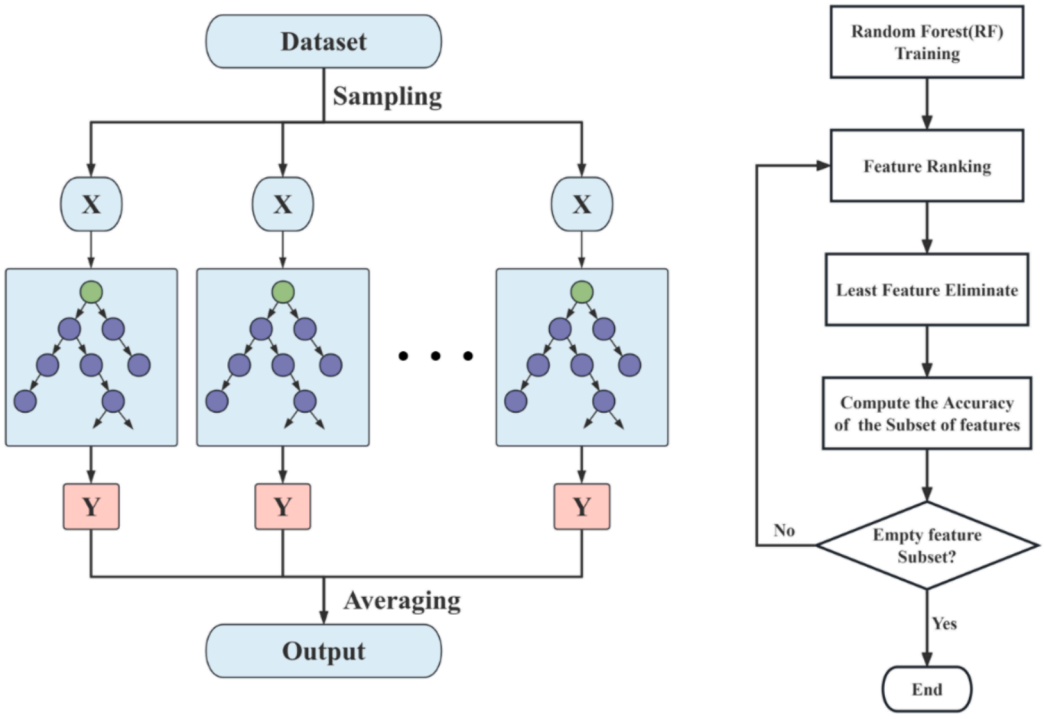


Fig. 6. Schematic Overview of Feature Selection and Model Optimization Using Random Forest and Recursive Feature Elimination with SHAP values. The process begins with the Random Forest algorithm evaluating feature importance and proceeds with iterative feature elimination based on SHAP values to refine the model and identify critical features for symbolic regression analysis.

$$\mathrm{Corr}(x_i, x_j) = \frac{\mathrm{Cov}(x_i, x_j)}{\sqrt{\mathrm{Var}(x_i)} + \sqrt{\mathrm{Var}(x_j)}} \tag{4}$$

where x_i and x_j are the variables under consideration.

Following the analysis of the PCC, the study further visualizes the relationships between the features through a heat map of the PCC. The heatmap in Fig. 5 offers a color-coded representation of the relative values, providing an intuitive visual display of the degree of linear correlation between pairs of features within the dataset. In this map, each cell represents the correlation between the features intersecting at that point, with a range of values from −1 to 1. A value of 1 signifies a perfect positive linear correlation, 0 indicates no linear correlation, and −1 implies a perfect negative linear correlation. This comprehensive view aids in the identification of potential relationships between the features, such as the notably high positive correlation between maximum axial strain and maximum torsional strain, as well as the negative correlations seen between variables such as Phase Difference and Shear Modulus of Elasticity. These insights are crucial for understanding the interactions between different mechanical properties and can guide further analyses and feature selection processes.

2.2.2. *Shapley Additive exPlanations*

Since PCC are limited to capturing linear relationships and cannot adequately represent the contribution of features to fatigue life prediction results, we implemented SHAP analysis to overcome this limitation. SHAP is particularly effective in providing a comprehensive understanding of how each feature contributes to the prediction, capturing both linear and non-linear relationships[52,53]. It calculates the contribution of each feature to a particular prediction, providing insights into the model's decision-making process. The SHAP value for an individual feature reflects its importance and impact on the output of the model, thereby offering an understanding of how features influence the final prediction outcome. This method facilitates transparent model interpretability, critical for validating the relevance and impact of the features used in the model. The calculation of SHAP values can be represented in a simplified form as follows:

$$\mathrm{SHAP}(x_i) = \sum_{S \subseteq F \setminus \{x_i\}} \frac{|S|!(|F| - |S| - 1)!}{|F|!} \left[f_S(x_i) - f_{S \setminus \{x_i\}}(x_i) \right] \tag{5}$$

where F is the set of all features, S is a subset of features excluding the feature x_i, f_S is the model prediction with features S, and $f_{S \setminus \{x_i\}}$ is the model prediction without the feature x_i. The formula calculates the average marginal contribution of a feature x_i across all possible combinations.

2.2.3. *Recursive feature elimination*

RF is a versatile machine learning method known for its robustness and accuracy. It operates by constructing a multitude of decision trees during the training phase and outputs the class that is the mode of the classes or mean prediction of the individual trees. This algorithm is esteemed for its proficient handling of high-dimensional datasets, where its ensemble approach integrates outcomes from numerous decision trees to mitigate overfitting while enhancing the generalization of the model. This methodology is conducive to identifying the underlying structure within complex datasets, enabling a robust feature analysis and extraction.

The RFE method[54], utilizing the RF algorithm enhanced by SHAP importance analysis, follows a systematic process depicted in Fig. 6. Initially, RF determines the importance of features, guided by SHAP values, to rank and progressively remove the least impactful ones. Post each removal, the model's accuracy is recalculated, allowing assessment of each feature's influence. This cycle continues until a refined subset of critical features remains, optimizing the model's performance and streamlining the feature set for effective symbolic regression in subsequent analysis stages.

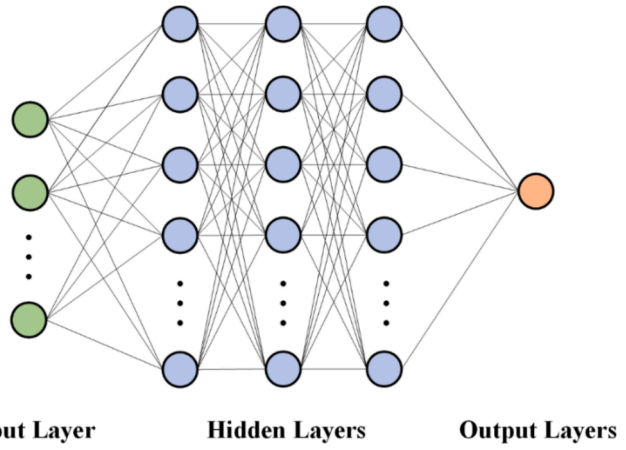


Fig. 7. Diagram of an Artificial Neural Network with Input, Hidden, and Output Layers. The network is designed to mimic the functioning of the brain, processing input data to produce an output.

2.3. Symbolic regression-neural network framework for multiaxial fatigue life prediction

This section introduces the proposed SR-NN method. Section 2.3.1 briefly introduces the fundamental concepts and architecture of the ANN model. Section 2.3.2 presents symbolic regression, which identifies optimal expressions for fatigue life prediction through key feature groups. Section 2.3.3 details the process of enhancing neural networks with these optimal expressions. Lastly, Section 2.3.4 offers a quantitative assessment and validation of the model's performance, ensuring its robustness and reliability in application.

2.3.1. Artificial neural network

ANN is a class of machine learning models inspired by the biological neural networks of animal brains. As Fig. 7 illustrates, ANN consists of an input layer that receives the data, hidden layers that process the data, and an output layer that delivers the final output. Each neuron in the hidden layers transforms its input via an activation function, enabling the network to capture complex patterns and relationships in the data.

2.3.2. Symbolic regression

Symbolic regression is a type of regression analysis that seeks to find an explicit mathematical model that describes the relationship between the dependent and independent variables in a dataset. Symbolic regression distinguishes itself from traditional regression analysis by not merely adjusting the parameters within a preset equation. Instead, it endeavors to discover the expression that most accurately encapsulates the observed data by searching through potential combinations of mathematical operations. Unlike conventional regression, which limits the model structure to a predefined form, symbolic regression is adept at identifying both the form and the parameters of the best-fitting mathematical models through the exploration of an unrestricted hypothesis space. This enables the uncovering of intrinsic, possibly complex nonlinear relationships within the data.

As illustrated in Fig. 8, genetic programming serves as one approach

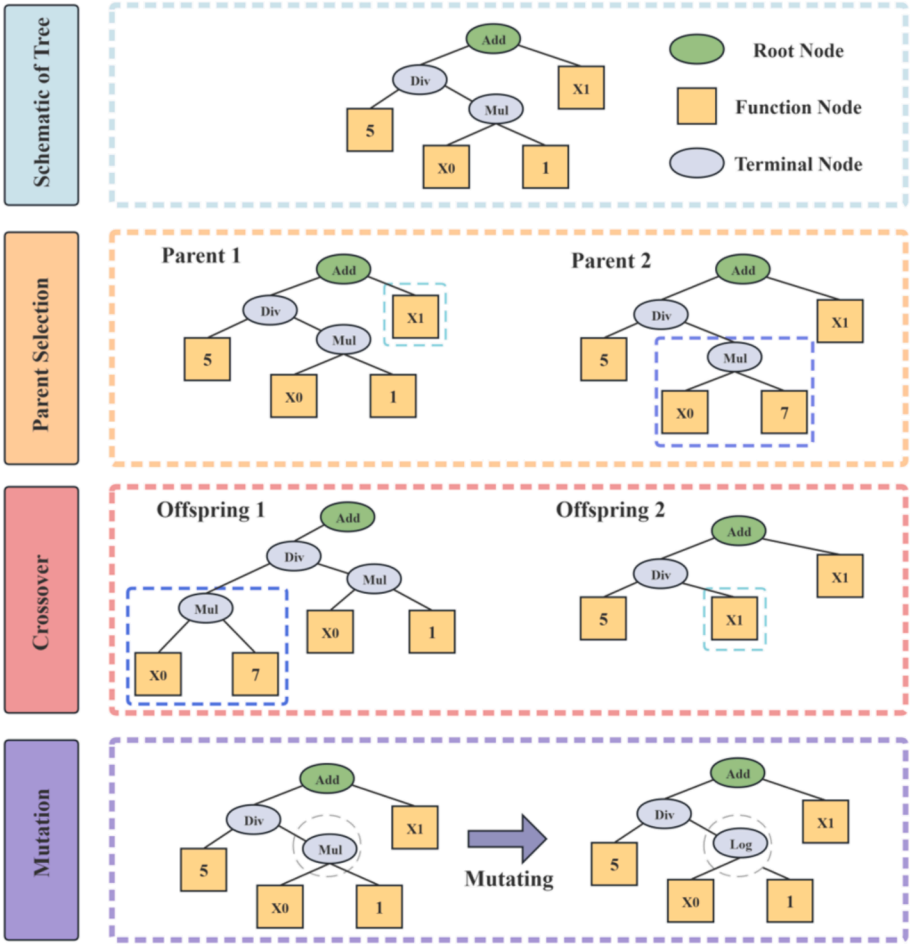


Fig. 8. Genetic Programming Process for Symbolic Regression. The diagram illustrates the initial tree structure, selection of parent expressions, and the generation of new offspring through crossover and mutation, highlighting the evolutionary approach to optimizing mathematical models.

Table 3
Genetic Programming Parameters for Symbolic Regression.

Parameter	Functions_set	Generations	Population_size	Metric	Crossover	Subtree_mutation	Hoist_mutation	Point_mutation	Max_samples
Value	+; −; *; /; lg	100	1000	MSE	0.7	0.1	0.05	0.1	0.9

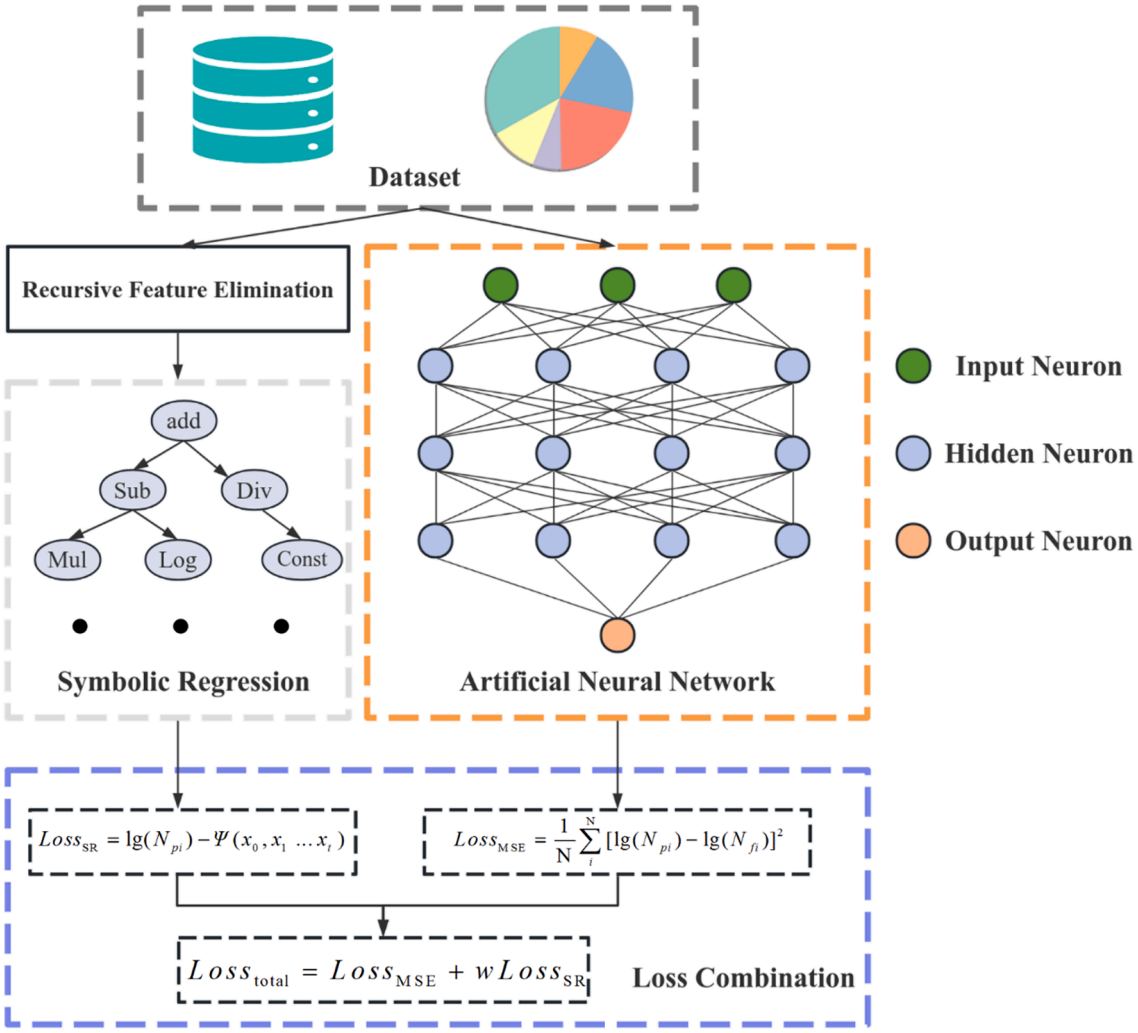


Fig. 9. Hybrid Predictive Framework Combining Neural Networks and Symbolic Regression. The model incorporates data through input neurons, processes it in hidden layers, and outputs results while optimizing a specialized loss function that integrates symbolic regression expressions and MSE to enhance prediction of fatigue life.

to implementing symbolic regression. It emulates the principles of natural selection to optimize mathematical expressions over successive generations. This methodology commences with the generation of an initial population of diverse symbolic expressions, conceptualized as trees composed of nodes representing various mathematical functions and operands. Through selection based on fitness criteria, the most promising trees are chosen to propagate into the next generation. Genetic operations, namely crossover and mutation, infuse the genetic programming algorithm with essential diversity. Crossover recombines portions of two parent trees to generate varied offspring, while mutation sporadically adjusts individual components within a tree, promoting a rich exploration of the solution space and fostering a broad spectrum of potential models.

Utilizing symbolic regression based on genetic programming, we conducted a 100-generation evolutionary process to optimize the feature group combinations for our model. The iterative process of genetic programming produces multiple candidate expressions with each generation, progressively refining them towards an optimal solution. We employ this technique to derive expressions that articulate the complex nonlinear relationships between the key feature group, determined by Recursive Feature Elimination, and fatigue life. This process was constrained to basic arithmetic operations: addition, subtraction, multiplication, division, and logarithms. For each generation, we generated a set of 1000 candidate combinations from which the 10 expressions with the highest fitness were selected. The selection was based on their ability to minimize the mean squared error (MSE) on a validation set. These top performers were used to seed the next generation. Fitness was assessed based on the precision and robustness of the expressions, ensuring that each successive generation honed in on more accurate and reliable predictive models.

This methodical approach allowed for the continuous improvement of the model, with the aim of arriving at an optimal combination expression that would enhance its predictive capabilities. For instance, we set the crossover probability to 0.7 to ensure a high rate of

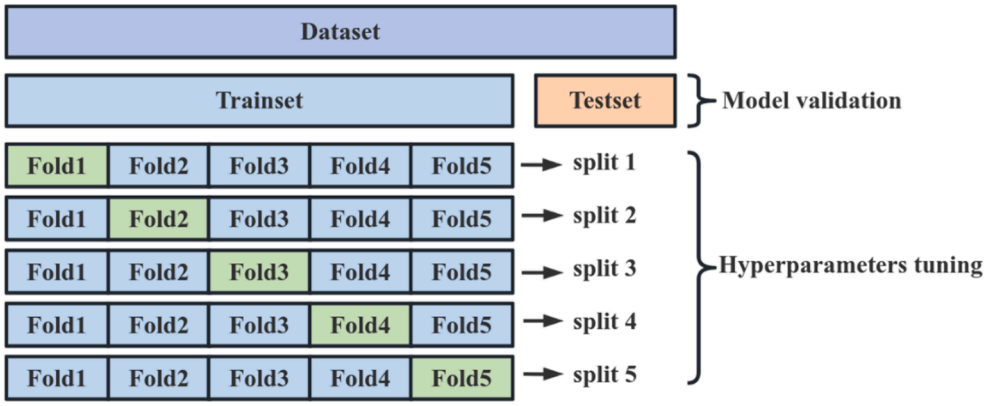


Fig. 10. Data Segmentation and Model Validation Workflow. The dataset is divided into training and test sets, with the training set undergoing 5-fold cross-validation to tune hyperparameters and ensure robust internal assessment.

recombination between candidate solutions, promoting diversity and preventing premature convergence. Similarly, a subtree mutation probability of 0.1 was used to introduce variations by randomly modifying parts of the expressions, which helps in exploring the solution space more thoroughly. Additionally, we implemented an adaptive probability scheme for genetic operations, dynamically adjusting crossover and mutation rates based on the current generation. This scheme gradually decreased the crossover probability while increasing mutation probabilities including subtree mutation, hoist mutation, and point mutation as the generations progressed. These mutation types allow for different scales of changes in the expressions, from small alterations to larger structural modifications. This adaptive approach aimed to balance exploration in early generations with exploitation in later ones, enhancing the algorithm's ability to find optimal solutions. Table 3 outlines the specific parameters used in the symbolic regression process, including the function set, population size, metric for evaluation, and genetic operation probabilities. This table provides a detailed overview of the genetic programming configuration used to derive the optimal symbolic expressions.

2.3.3. Symbolic regression-neural network framework

As illustrated in Fig. 9, we introduce a new predictive framework that synergizes the capabilities of neural networks with symbolic regression. Beginning with input features entering the input layer, the network engages an activation process with initial weights and biases set to random values. These parameters are then iteratively refined through the training process using gradient descent to minimize output error. The unique aspect of this framework lies in its specialized loss function design, which harmoniously integrates the symbolic regression-generated mathematical expressions with the traditional loss-mean squared error (MSE). This fusion facilitates the detailed unraveling of complex mathematical relationships between input features and fatigue life, further transposed into a novel loss term that couples the quantitative life values with computational elements derived from a synergistic feature combination.

Thus, we propose the following loss function:

$$Loss_{\text{total}} = Loss_{\text{MSE}} + wLoss_{\text{SR}} \tag{6}$$

$$Loss_{\text{MSE}} = \frac{1}{\text{N}} \sum_{i}^{\text{N}} \left[\lg(N_{pi}) - \lg(N_{fi})\right]^2 \tag{7}$$

$$Loss_{\text{SR}} = \lg(N_{pi}) - \Psi(x_0, x_1 \ldots x_t) \tag{8}$$

where N_{pi} is the predicted fatigue life, $\Psi(x_0, x_1 \ldots x_t)$ is the symbolic regression function of input features, N_{fi} is the experimental fatigue life, and N is the size of training dataset, t represents the number of key feature group.

Within the neural network, the output of each iteration, which predicts fatigue life, is used to subtract from the target value derived from the feature combination term. This difference is utilized as a new loss term. Notably, the target value in this context remains constant across iterations, based on the expression obtained from symbolic regression, while the predicted value is continually updated through the training of the neural network.

The integration of these two aspects, symbolic regression and neural network optimization, allows for a hybrid-approach in model training. On one hand, the network learns to minimize the MSE loss. On the other, it concurrently minimizes the discrepancy between its predictions and the expression provided by symbolic regression, effectively ensuring that the model adheres to the underlying physics captured by the symbolic regression as it optimizes the predictions. The SR loss specifically excavates the underlying physical principles governing the data, embodying a deeper layer of the data's intrinsic patterns that traditional loss functions in neural networks might overlook. It results in a model that is inherently more interpretable, as the symbolic expressions are human-readable and convey meaningful scientific insights. Moreover, by grounding the network's predictions in the physical laws inferred from the data, it promotes a form of learning that is more generalizable and resilient to overfitting, thus offering substantial improvements over the conventional neural network training methodologies that rely solely on data-driven loss minimization.

2.3.4. Model evaluation criteria

To accurately assess the predictive performance of the proposed fatigue life prediction framework, R-squared and MSE have been chosen as the principal metrics. R-squared, also referred to as the coefficient of determination, quantifies the amount of variance in the observed data that can be explained by the model's input variables. MSE provides an assessment of the average squared deviation between the model's predictions and the observed values, offering an aggregate measure of the model's precision.

In the application of machine learning methodologies, it is a common practice to partition the dataset into separate training and testing sets. For this paper, stratified sampling was utilized to constitute a discrete test set for model validation, extracting 15 % from each of the six different material datasets. The remaining data formed a substantial training set, which was subjected to 5-fold cross-validation for internal assessment. This foundation facilitated the employment of grid search techniques aimed at identifying the optimal hyperparameter combination. The schematic representation of this data segmentation and validation process is captured in Fig. 10. In our comparative analysis, we evaluated several machine learning models, including SR-NN, ANN, SVM[55], RF[56], and XGBoost[57]. The hyperparameters for each

Table 4
Hyperparameter Search Space for Various Machine Learning Models.

Model	Hyperparameters
SR-NN	layer; layer_size; alpha; learning rate; activate function; loss weight
ANN	layer; layer_size; alpha; learning rate; activate function
SVM	svr_C; svr_kernel; svr_degree; svr_gamma
RF	n_estimators; max_depth; min_samples_split; min_samples_leaf
XGBoost	n_estimators; learning rate; max_depth; subsample; colsample bytree

model, fine-tuned through grid search, is detailed in Table 4. For model validation, we adopted the mean R-squared (mR-squared) metric, representing the average R-squared value obtained across the *k*-folds of the training set, to measure the model's predictive performance.

$$\text{mR - squared} = \frac{1}{k}\sum_{i=1}^{k} R_i^2 \tag{9}$$

where k is the number of folds in the cross-validation, R_i^2 is the R-squared value for the i-th fold.

3. Results and discussions

3.1. Feature analysis and key feature selection

3.1.1. Visualizing multidimensional data and feature importance

In our statistical analyses, the Parallel Coordinates Plot (PCP) is utilized as an advanced visualization technique to examine multidimensional data structures within a two-dimensional space. As depicted in Fig. 11(a), each vertical axis represents a unique feature, and individual observations are rendered as polylines intersecting these axes. This method elucidates patterns and potential correlations across various features, as evidenced by the spread and clustering of lines, indicative of variability and inter-features relationships respectively.

Expanding on these multi-dimensional insights, Fig. 11(b) and 10(c) present the SHAP analysis conducted on the original dataset. Fig. 11(b) reveals the average SHAP values for each feature through a bar plot, with longer bars indicating a more pronounced effect on the model's output. This average impact is further detailed in Fig. 11(c)'s beeswarm plot, which provides a granular view of the SHAP values' distribution for each feature, reflecting not just the mean effect but also its variation within the data. Collectively, these SHAP visualizations emphasize the contributions of individual features to the predictive model, offering a comprehensive understanding of their influence on the results.

3.1.2. Dimensionality reduction and feature selection

Confronted with the high dimensionality of our dataset, as evident from the PCP, the necessity to refine the feature space is critical. An overabundance of features not only obfuscates discernible patterns in prediction but also imposes a burden on the process of symbolic regression. It tends to yield unwieldy and extensive symbolic expressions that are computationally intensive and challenge interpretability, thus diluting the core advantage of symbolic regression — the generation of concise models that are as comprehensible as they are functional,

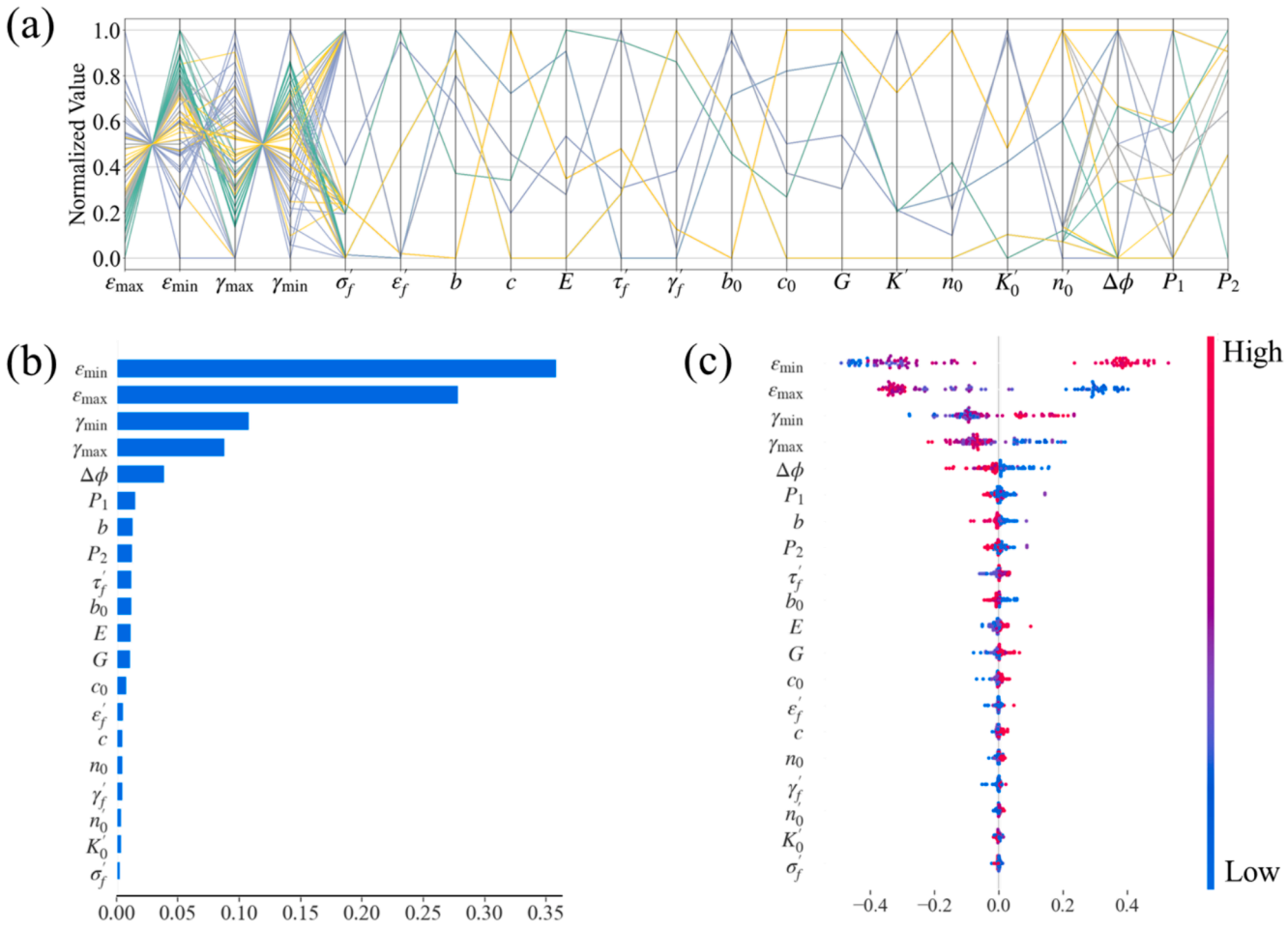


Fig. 11. Multidimensional Data Analysis and SHAP Value Visualization. (a) Parallel Coordinates Plot illustrating inter-feature relationships and variability. (b) Bar plot of average SHAP values indicating feature impact on model output. (c) Beeswarm plot detailing the distribution of SHAP values, offering insights into feature contribution and variation within the predictive model.

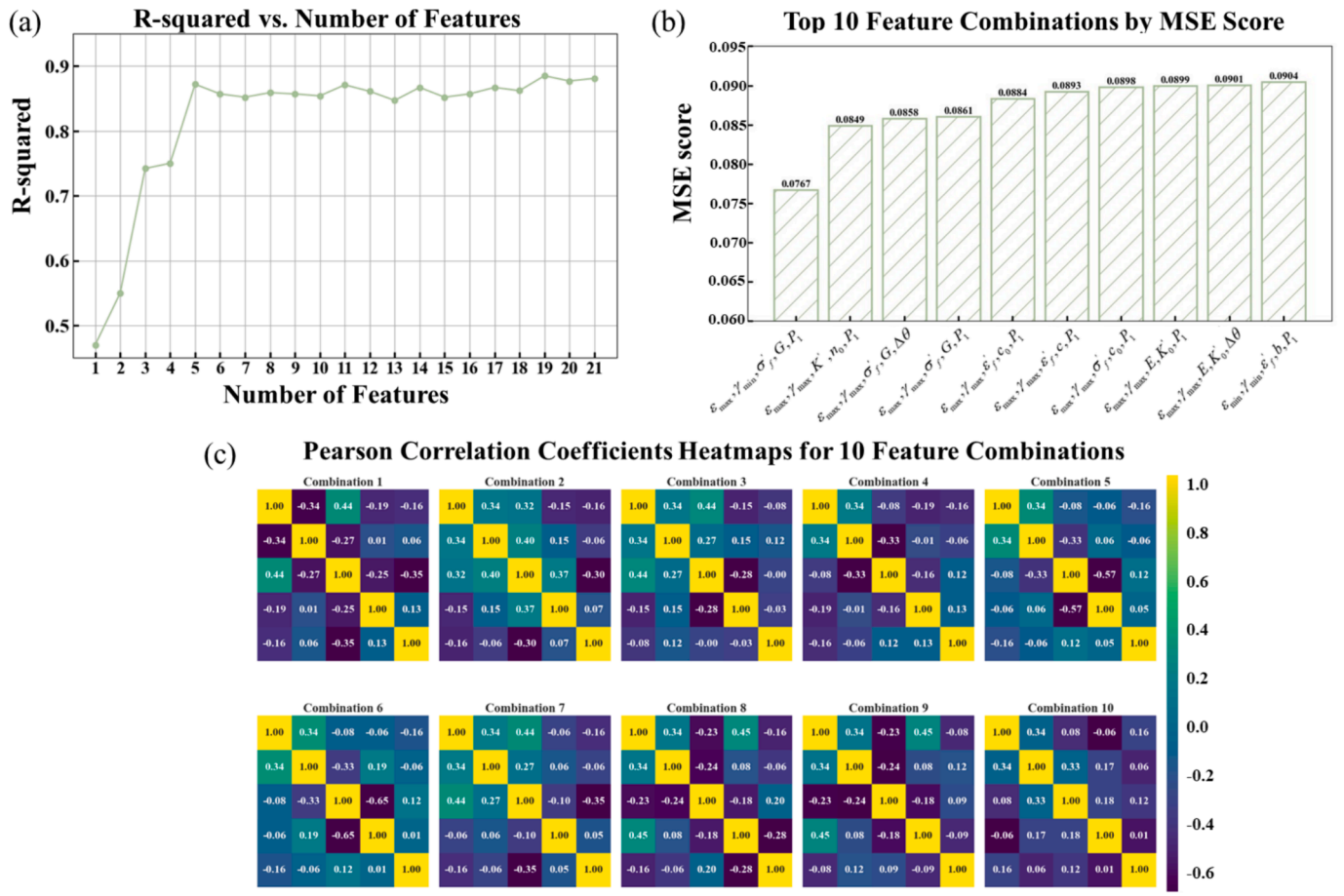


Fig. 12. Evaluation of Feature Selection and Interdependence. (a) depicts the R-squared values as a function of the number of features, highlighting the point of diminishing returns in model performance. (b) shows a ranking of the top 10 feature combinations by MSE score, indicating a balance between model accuracy and complexity. (c) displays the Pearson Correlation Coefficients Heatmaps for the 10 feature combinations, showcasing the relative independence of selected features critical for model simplicity and predictive accuracy.

especially when integrated with neural network frameworks. Consequently, RFE guided by SHAP values transcends mere dimensionality reduction; it is a strategic approach to alleviate the detrimental effects of feature complexity on symbolic regression. Its methodical feature selection harmonizes model precision with conciseness, fostering the derivation of symbolic equations that excel in both accuracy and intelligibility. Fig. 11(a) demonstrates the R-squared score versus the number of features, illustrating how model explanatory power reaches a balance with a subset of five features, indicating the efficacy of RFE in identifying an optimal subset of predictors. Thus, we have determined the

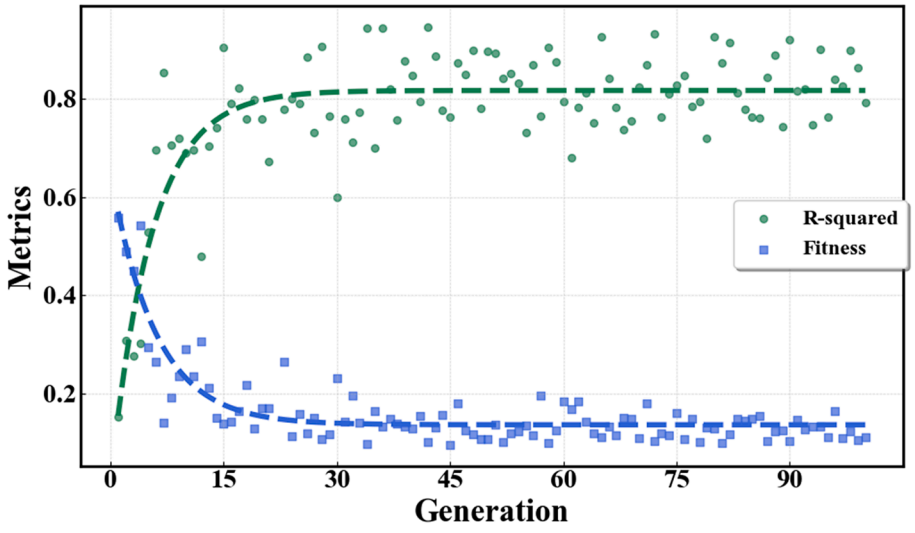


Fig. 13. Evolution of Model Performance across Generations. The graph shows the progression of R-squared and fitness metrics over 100 generations, highlighting the algorithm's rapid initial improvement followed by stabilization, reflecting the effective optimization in the symbolic regression process.

Table 5
Selected Symbolic Expressions with Corresponding R-squared and Fitness Values.

No.	Expression	R-squared	Fitness
case1	$\lg((\lg(\varepsilon_{max}-\gamma_{min}))/(-0.23)+(\lg\varepsilon_{max})/(-0.23)+(2\lg\gamma_{min})/(-0.514\gamma_{min}^{2}))+\lg G)-\varepsilon_{max}$	0.834	0.112
case2	$\lg((\lg(\varepsilon_{max}+0.1)/\gamma_{min}^{3}))+(\lg(0.277+\varepsilon_{max})/\lg(0.928))+((\lg G(\varepsilon_{max}+0.1)(-0.396))+\lg(\sigma_f'))$	0.851	0.105
case3	$\lg G+(\gamma_{min}-((((\varepsilon_{max}-0.205)-0.474\varepsilon_{max})-0.205)+P_1\varepsilon_{max})-\lg(\sigma_f'))$	0.814	0.119

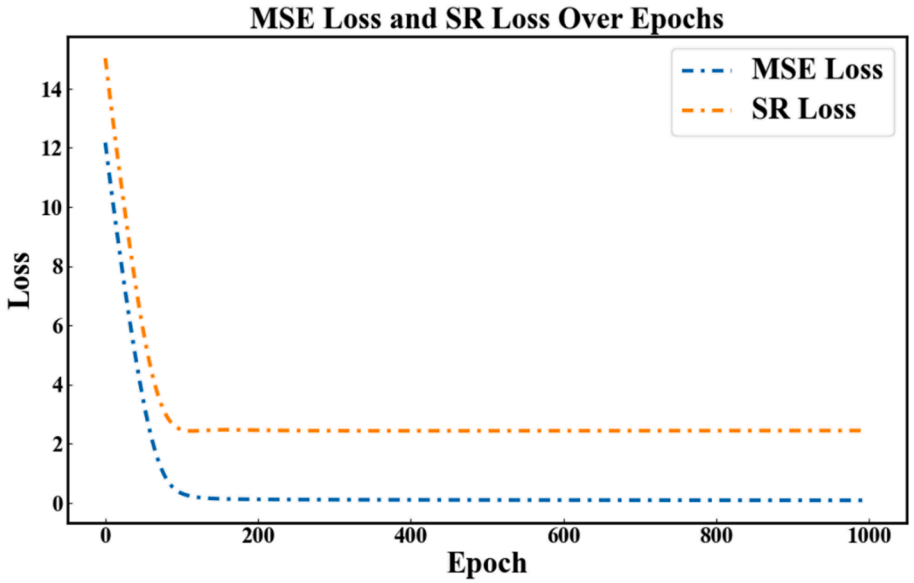


Fig. 14. Training Progression of the Proposed Model Losses over Epochs. The graph tracks the decrease in MSE Loss (blue) and SR Loss (orange) across epochs, demonstrating the model's improvement in accuracy and adherence to the underlying physical principles.

number for our key feature group to be five. Fig. 12(b) ranks the top 10 features combinations by their MSE scores, revealing the most effective feature set that balances model accuracy and complexity.

Following the identification of this key feature group, we employed PCC analysis to ascertain the interdependencies within the selected features. The heatmaps in Fig. 12(c) demonstrate that our chosen features—Maximum Axial Strain, Maximum Torsional Strain, Fatigue Strength Coefficient, Shear Modulus of Elasticity, and Path-related Parameter P_1—offer complementary information to the model, reflecting their independence. This ensemble of key features encompasses crucial aspects: load characteristics, mechanical properties, and loading path, thereby laying the groundwork for the construction of a succinct and interpretable expression in symbolic regression, paving the way for enhanced predictive accuracy and model simplicity.

3.2. Fatigue life prediction based on the proposed SR-NN framework

3.2.1. Optimization results of symbolic regression on key feature group

Fig. 13 delineates the evolutionary progression of two pivotal metrics, R-squared and fitness, over the course of 100 generations within the symbolic regression framework. The fitness trajectory, illustrated by a dashed blue line, shows a precipitous rise in the nascent stages of the evolutionary process. This upsurge is emblematic of the algorithm's rapid convergence on more optimal solutions. Subsequent to this initial leap, the metric levels off, with minor undulations indicative of a phase where exploration within the solution space has reached a form of equilibrium. In parallel, the R-squared metric, rendered as a solid green line, experiences a similar early ascent, swiftly achieving a high steady state, which it consistently maintains near 0.8. Such an elevated R-squared value underscores a substantial explanatory capacity of the model, intimating that the evolved symbolic expression possesses a strong correlative relationship with the observed data. With a focus on maintaining a balance between simplicity of the expressions and their performance, as illustrated by an R-squared above 0.8 and a fitness below 0.15, we selected three cases for further investigation, as shown in Table 5.

3.2.2. Evolution of performance metrics during training

After integrating the symbolic expressions derived from genetic programming into the loss function of a neural network, we monitored the evolution of two parts of the loss function throughout the training process, as depicted in Fig. 14. As the training progresses, both the MSE Loss and the SR Loss exhibit a substantial decrease. The MSE Loss, shown in blue, and the SR Loss, in orange, simultaneously decline, reflecting the model's concurrent learning from both components. This simultaneous reduction indicates that the model benefits from the MSE Loss's focus on prediction accuracy and the SR Loss's emphasis on the data's underlying physical laws. The SR Loss ensures that the model continues to refine its predictions in alignment with the essential principles, complementing the MSE Loss and enriching the learning process. Therefore, even as the losses reach their respective minima, the model's learning is sustained through this integrated approach, promoting a comprehensive understanding and enhancing predictive performance.

Despite both loss components reaching their minima in a relatively synchronous manner, as illustrated, the SR Loss's ability to capture the intrinsic properties of the dataset enables the model to continue its optimization beyond the point where traditional loss metrics might suggest that learning has stalled. This hybrid-loss approach ensures that our model not only fits the data well but also adheres closely to the fundamental behaviors it aims to represent, leading to improved generalization and robustness in predictions.

3.2.3. Evaluating SR-NN predictive performance

Fig. 15 illustrates the predictive performance of different symbolic

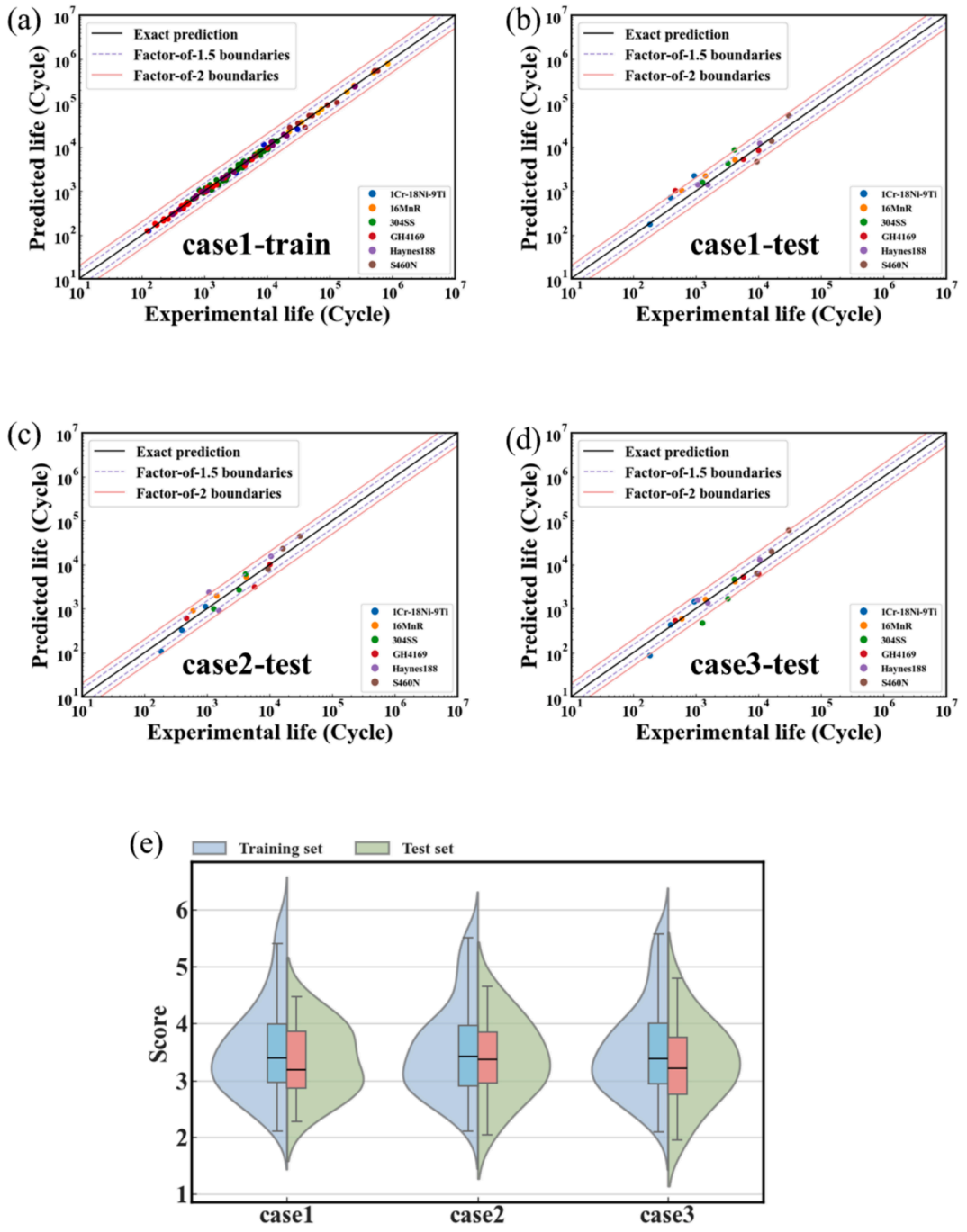


Fig. 15. Comparison of Predictive Accuracy for Different Symbolic Regression Cases. (a) and (b) display the predictions for case1 on the training and test sets, respectively, with a high density of points within the factor-of-1.5 boundaries indicating a precise fit. (c) and (d) depict the results for cases 2 and 3 on the test sets, with case3 showing predictions consistently within the factor-of-2 boundaries, highlighting its robustness across various conditions. (e) illustrates the score distributions for the training and test sets across the three cases, indicating the model's performance and generalization capability.

Table 6
Optimal Hyperparameter Selection and Cross-Validation Performance.

No.	layer_size	alpha	learning_rate	activate_function	loss_weight	mR-squared
case1	{32,128,32}	0.01	0.001	sigmod	1.0	0.920
case2	{32,64,32}	0.001	0.01	relu	1.5	0.934
case3	{32,64,32}	0.01	0.005	relu	1.0	0.911

regression expressions guiding a neural network for three separate cases. Table 6 presents the selected parameters for each of these cases along with the mean R-squared values obtained from 5-fold cross-validation, highlighting the tailored approach to optimizing model architecture and learning dynamics for each scenario. Fig. 15(a) presents the results of case1 on the training set, showing a high concentration of data points

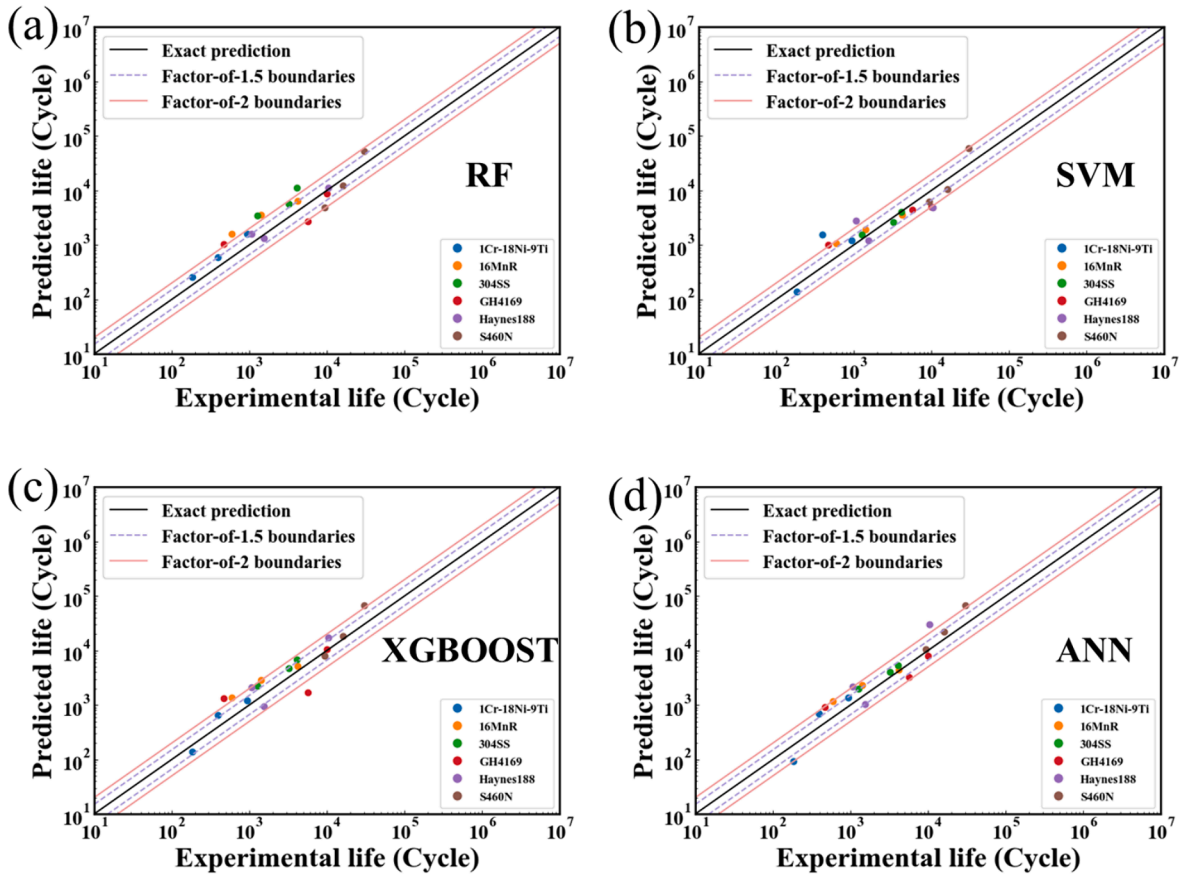


Fig. 16. Evaluative Prediction for Various Predictive Models. (a)~(d) contrast the predicted life cycles with experimental data, delineating precision within specified error boundaries for multiple modeling approaches.

within the factor-of-1.5 boundaries, indicating an excellent fit to the exact predictions. This level of fit is further exemplified by the mean R-squared value of 0.920 for case1, as outlined in Table 6, demonstrating the model's strong predictive power. This performance is mirrored in the test set results displayed in Fig. 15(b), suggesting that the model trained with the case1 expression generalizes well to unseen data. In contrast, Fig. 15(c) and Fig. 15(d) display the results for cases 2 and 3 on their respective test sets. While case2 achieves the highest mean R-squared value of 0.934, indicating superior accuracy, case3 shows a consistent distribution of data points within the factor-of-2 error bounds on the test set. Despite case3 having a slightly lower mean R-squared value of 0.911, it has more points falling on the exact prediction line. Fig. 15(e) displays violin plots of the score distributions for the training and test sets across the three cases. The Score represents the error between the predicted and actual fatigue life values. The width of the violin plots indicates the density of data points at different score levels, with wider sections representing a higher concentration of points. This visualization highlights the model's performance consistency across different datasets, showing similar score distributions for both training and test sets. The narrow width of the plots at low error levels indicates that the majority of predictions are close to the actual values, reflecting the model's robust predictive performance.

3.3. Performance evaluation

Fig. 16 provides a comparative analysis of the predictive models using various machine learning algorithms. Table 7 details the hyperparameter configurations and the mean R-squared values from cross-validation for the traditional machine learning models, including RF, SVM, XGBoost, and ANN. The table demonstrates how each model's hyperparameters are tuned to optimize performance, with the ANN model achieving the highest mean R-squared value among the traditional algorithms. Fig. 16(a) ~ (d) show the predicted versus experimental life cycles for the different models, with exact predictions and boundaries for factor-of-1.5 and factor-of-2 for different cases. By comparing the results with those in Fig. 15, the SR-NN model shows substantial improvements in predictive accuracy, with a 55.4 % reduction in MSE compared to the RF model and 45 % lower than the SVM model. These enhancements validate the integration of symbolic regression into the neural network framework for complex predictive tasks. Fig. 17(a) outlines these performance metrics, presents a side-by-side comparison of MSE and R-squared values across the models, where the proposed SR-NN model exhibits superior accuracy and generalization capabilities over its counterparts, while Fig. 17(b) employs violin plots to display the distribution of model scores, highlighting the comparative performance on the training and test sets and thereby

Table 7
Hyperparameter Tuning and Cross-Validation Performance of Traditional Models.

Model	Hyperparameters	mR-squared
RF	max_depth: 10; min_samples_leaf: 1; min_samples_split: 2; n_estimators: 300	0.862
SVM	svr_C: 10; svr_degree: 2; svr_gamma: 'scale'; svr_kernel: 'rbf'	0.838
XGBoost	colsample_bytree: 0.9; learning_rate: 0.1; max_depth: 3; n_estimators: 300; subsample: 0.8	0.896
ANN	Layer_size:{128,128,128}; learning rate:0.001; loss function: 'MSE';	0.903

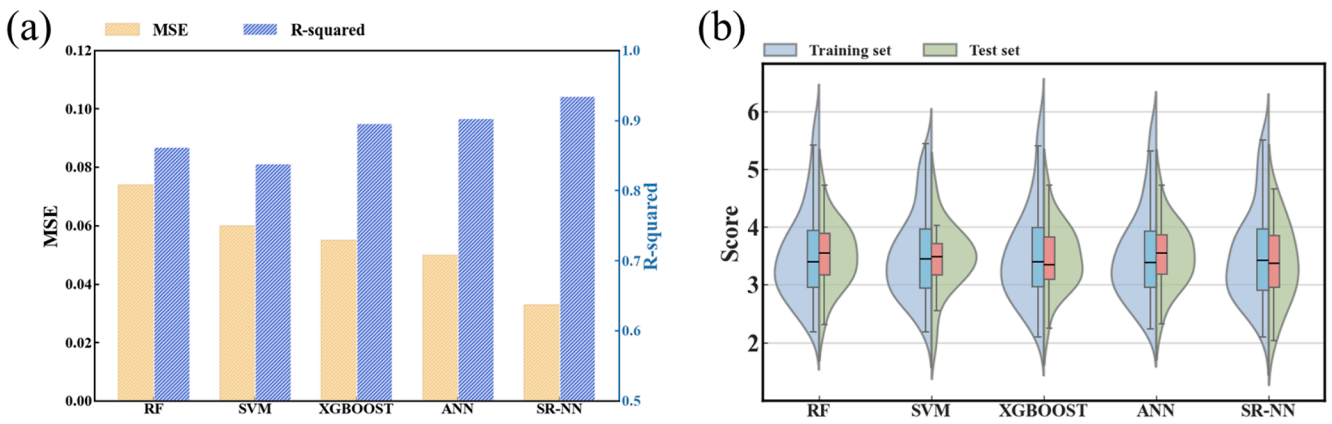


Fig. 17. Performance Metrics and Distribution across Models. (a) exhibits a comparison of models using MSE and R-squared values, showcasing the superior accuracy and generalization capabilities of the SR-NN model over other models such as RF, SVM, XGBoost, and ANN. (b) illustrates the distribution of performance scores for each model, comparing results on training and test sets.

underlining the enhanced reliability and robustness of the proposed model.

In light of the above analysis, it is demonstrated that our proposed SR-NN model, which synergistically combines symbolic regression with neural network, achieves a level of predictive precision that surpasses that of traditional models. This hybrid approach transcends the confines of traditional loss minimization, ensuring that the predictive model not only aligns with the MSE criterion but also complies with the fundamental characteristics inherent in the dataset. Such an integration facilitates a dual-objective optimization where the neural network is trained to concurrently minimize the conventional data-driven loss and the difference illuminated by symbolic regression expressions. Through the uncovering of data's underlying intrinsic relationships via symbolic regression, the SR-NN model harnesses a deeper comprehension of the data's intrinsic patterns—patterns that may be overlooked by conventional machine learning algorithms. The innovative loss function crafted for this model encapsulates both mathematical precision and data fidelity, leading to predictions that are not only accurate but also provide insights into the training process and the governing patterns of critical features. This fosters a learning process rooted more deeply in the principles that govern the underlying data structures under study, substantially reducing the risk of overfitting while bolstering the model's reliability and applicability. By uncovering the underlying relationships in the dataset, SR-NN models are a significant improvement over traditional training methods that rely solely on data-centric loss minimization. The success of this model underlines the potential of such an approach to yield not merely predictive accuracy, but also to provide a gateway to deeper scientific insights and more robust generalizations across varied datasets.

4. Concluding remarks

In this study, we meticulously curated key features from a comprehensive dataset of multiaxial fatigue experiments. Utilizing symbolic regression, we extracted expression that encapsulate the essence of fatigue life predicted on the basis of these salient features. These formulae have been integrated into the loss function, in conjunction with the conventional MSE component, to craft a customized loss construct. The model underwent rigorous validation against a stratified sample from the test set, carefully selected to reflect the diversity and balance of the dataset. We benchmarked the model's predictive prowess and operational efficiency against established conventional models and standard neural networks, demonstrating its superior performance and efficacy. The main findings are summarized as follow:

(1) We proposed a novel Symbolic Regression-Neural Network (SR-NN) framework that utilizes a hybrid loss function, combining traditional MSE with insights derived from symbolic regression. This approach not only explicates the underlying physical relationships within the multiaxial fatigue life data but also leverages this understanding to significantly enhance the model's predictive accuracy on the established database.
(2) We employed SHAP-informed Recursive Feature Elimination to discern key feature groups, effectively reducing dimensionality and refining the predictive feature set. This strategic feature selection process not only enhances the efficiency of our model but also provides critical insights into selecting important attributes for new materials, guiding the development of more targeted and effective fatigue life prediction models.
(3) Compared to traditional models, our SR-NN framework demonstrated a notable enhancement in accuracy, with error rates 55.4 % lower than the RF model and 45 % lower than the SVM model. This significant improvement in predictive precision is credited to our model's unique ability to integrate and apply the underlying physical relationships revealed through symbolic regression, distinguishing it from conventional predictive approaches.

Traditional empirical formulas based on extensive experimental datasets for conventional materials have long been the foundation for fatigue life predictions. However, in the current era of rapidly evolving materials science, there is a need for complementary approaches that offer greater flexibility and adaptability. Our method aims to address this need by providing an approach that can handle diverse materials without requiring separate physical models for each type, and excels in incorporating new data to refine predictive capabilities. By initially selecting key features and subsequently utilizing the unique dataset, our proposed framework seeks to uncover underlying physical relationships associated with the material of interest, potentially guiding the training of machine learning methods. This approach may provide additional insights that could enhance the accuracy and reliability of fatigue life predictions, offering a useful complement to traditional empirical formulas.

Last but not least, our method shows significant potential for application to emerging materials while also advancing the interpretability of machine learning in fatigue analysis. As the field of materials science continues to evolve rapidly, with new materials being developed at an unprecedented rate, our approach offers a novel solution that combines the power of data-driven methods with enhanced transparency. By integrating symbolic regression-derived expressions into the neural network architecture, we provide a framework for more targeted and interpretable predictions, well-suited to the rapid development of new

materials. This strategy adapts to the dynamics of material innovation, allowing for efficient analysis of fatigue behavior in materials with limited historical data or unique properties, while simultaneously offering insights into the underlying relationships governing fatigue life. Consequently, this approach represents a promising tool for researchers and engineers at the vanguard of materials science and fatigue analysis, particularly in addressing the challenges posed by novel materials and complex loading scenarios.

CRediT authorship contribution statement

Peng Zhang: Writing – original draft, Visualization, Validation, Software, Methodology, Investigation, Formal analysis, Data curation. **Keke Tang:** Writing – review & editing, Supervision, Resources, Project administration, Methodology, Funding acquisition, Conceptualization. **Anbin Wang:** Formal analysis, Data curation. **Hao Wu:** Writing – review & editing, Methodology, Funding acquisition. **Zheng Zhong:** Writing – review & editing, Supervision, Project administration, Conceptualization.

Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

Data availability

Data will be made available on request.

Acknowledgement

The work was funded by National Natural Science Foundation of China (11872278, 12372081), Science and Technology Commission of Shanghai Municipality (21ZR1467200), and the Fundamental Research Funds for the Central Universities.

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